

Advocate Guide: Qiskit v2.X Certification

Overview

We are excited to share that a new Qiskit Certification is on the way and will be released soon!

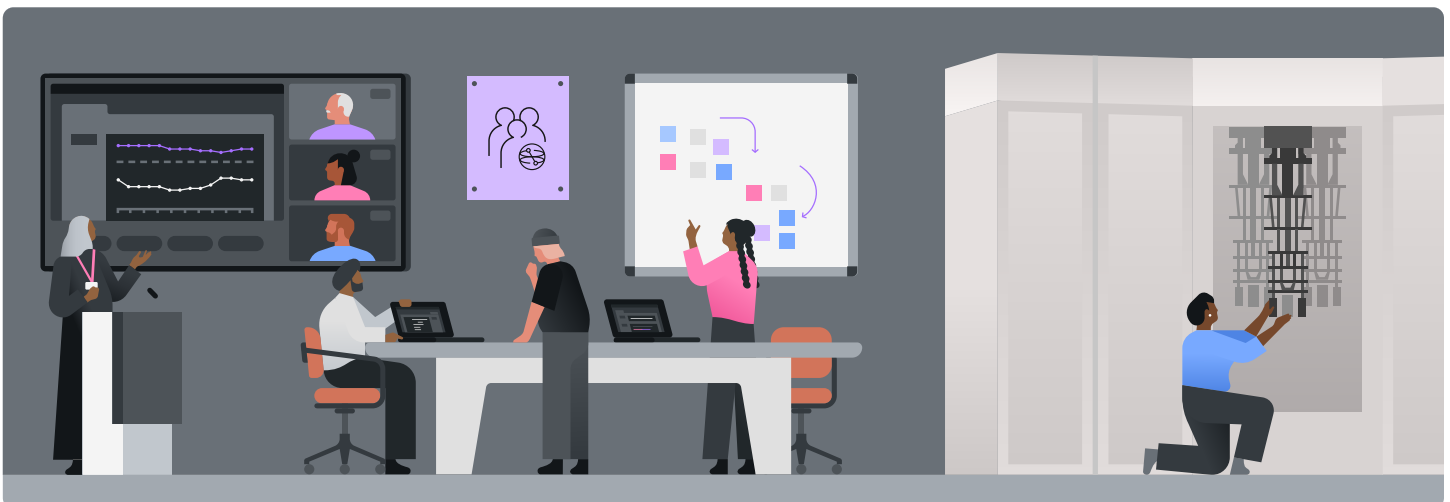
The Qiskit v2.X Certification is designed to validate your expertise in quantum computing using the latest in Qiskit, and by preparing for the associated examination, you will not only enhance your knowledge but also demonstrate your sustained proficiency in quantum programming.

As valued Qiskit Advocates, we want to provide you with a sneak peak of the Qiskit v2.X Certification exam, so that you can start preparing for this important certification.

On the following page you will find a syllabus and list of the topics covered in the Qiskit v2.X Certification exam. Alongside the topics, we have prepared a list of helpful resources you may use to prepare.

As you go through the topics and resources, please feel free to use the Qiskit advocate hub on Discord to discuss concepts with, ask questions to, and get help from your peers. You may use the [Qiskit forum](#) to do so.

We look forward to having you be among the first to obtain the new Qiskit v2.X Certification and appreciate your ongoing dedication and contributions to the Qiskit community.



Qiskit v2.X Certification Exam Topics	Resources
Perform quantum operations - Define Pauli Operators - Apply quantum operations	Documentation: Operators overview Documentation: Operator class
Visualize quantum circuits, measurements, and states - Visualize quantum circuits - Visualize quantum measurements - Visualize quantum states	Documentation: Visualize circuits Documentation: Visualize results Documentation: Plot quantum states
Create quantum circuits - Construct basic quantum circuits - Construct dynamic circuits - Construct parameterized circuits - Transpile and optimize circuits	Documentation: Construct circuits Documentation: Circuit library Documentation: Classical feedforward and control flow Documentation: Parameterized circuits Video: Introduction to transpilation Documentation: Introduction to Transpilation Documentation: Optimize for hardware Documentation: Set transpiler optimization level
Run quantum circuits - Differentiate execution modes to optimize job queuing - Run quantum circuits with a real hardware in the IBM Runtime provider	Documentation: Execution modes Documentation: Execute on hardware Documentation: Get started with primitives
Use the sampler primitive - Set sampler primitive options such as resilience levels - Bypass runtime error mitigations and implement your own - Understand the theoretical background behind the sampler primitive	Video: An Introduction to Qiskit Runtime Primitives V2 Documentation: Sampler options
Use the estimator primitive - Set estimator primitive options such as resilience levels - Understand the theoretical background behind the estimator primitive	Video: An Introduction to Qiskit Runtime Primitives V2 Documentation: Estimator options
Retrieve and analyze the results of quantum circuits - Retrieve previous experiment results (session/runtime) - Monitor jobs	Documentation: Save and retrieve jobs Documentation: Monitor or cancel a job
Operate with OpenQASM - Structure types in OpenQASM 3 programs - Interpret OpenQASM semantics for versions 2 and 3 - Interoperate different versions of OpenQASM with Qiskit - Interact with the Qiskit IBM Runtime REST API	Documentation: Introduction to QASM OpenQASM types and casting OpenQASM syntax Documentation: OpenQASM 3 and the Qiskit SDK Documentation: OpenQASM 2 and the Qiskit SDK Documentation: Qiskit Runtime REST API